

Emerging Applications

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Emerging Applications



HOME
SOLUTIONS



ENTERPRISE
SOLUTIONS



MOBILE
SOLUTIONS



EMBEDDED
SOLUTIONS

E M E R G I N G A P P L I C A T I O N S

Outline

■ Emerging applications for ARM IP

- Healthcare and fitness
- Electronic toys
- MCU-class networking



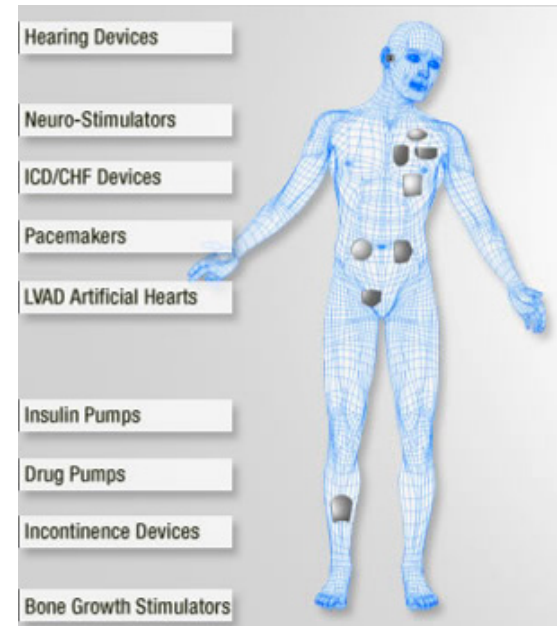
■ Channels for easy IP access

- Access™ Library program
- DesignStart™ program
- Foundry program



■ Ready for globalisation

- New customers
- New markets



■ Conclusion

Your Partner for Emerging Applications

- World-class technology portfolio in breadth and depth
 - Pioneering business model
 - Solid, growing market position
 - Broad range of applications
 - Significant investment to ensure relevance
- Change is accelerating, continued success requires pushing the envelope further
 - Need to go beyond the next generation; get “outside of the box”
 - Growth through new trends, obstacles, opportunities
 - Leverage cumulative investment to efficiently maximise ROI
- ARM is the ideal partner for Emerging applications
 - Strong position, relevant portfolio, long-term technology commitment



Healthcare and Fitness

- Largest opportunity to add value in the world
 - Healthcare systems under extreme stress despite \$3T global spend
 - Improvements needed:
In efficiency, reliability, privacy, quality of life
 - Product opportunities:
For prevention, monitoring, therapy, services, IT
- Emergence and growth of telemedicine
 - Removes distance between provider and patient
 - Need for connectivity increases hardware and software integration challenges
 - RealView[®] Development Suite 3.0 includes real-time system models, SoC Designer interoperability, CoreSight[™] debug support



Healthcare and Fitness

- Connectivity of monitoring equipment enables patient dashboard, new services



- Data aggregation aids monitoring, analysis, efficiency
- Cortex™-M3 processor provides performance headroom, ARM portfolio future-proofs roadmap
- RealView Microcontroller Development Kit includes RTOS, TCP/IP stack, mini web server, Flash file system

- Regulatory requirements for safety and effectiveness

- ARM IP delivers field-proven reliability, robustness
- Safety in numbers: billions of ARM processors and Physical IP components in production
- A small step towards substantial equivalence



Healthcare and Fitness

- Portability and battery life key differentiators

- Integration a necessity → AMBA[®] interface
- Metro[™] low-power platform boosts battery life
- Intelligent Energy Manager technology cuts CPU power consumption



- Privacy and security provisions of HIPAA driving complexity



- TrustZone[®] technology isolates sensitive data and applications while cryptography raises performance requirements
- AMBA 3 AXI[™] system-level support of TrustZone technology

- Fitness and prevention funding innovation

- HRM is a \$200M+ industry already
- New sensors, new connectivity → new capabilities



- HRM
- Altimeter
- Barometer
- Thermometer
- Compass
- GPS
- Nat'l Geo. TOPO
- USB
- Statistics, analysis
- Coaching
- Date and time

Electronic Toys

- Electronic toys comprise well over 40% of total toy sales
- Children's expectations are skyrocketing!
 - Interactivity
 - Connectivity
 - Gaming
 - Content
 - Multiple functions
 - Speech recognition
 - Servo motors
 - Sensors
 - Programmability
 - Personality
- Yesterday's expensive, high-tech product is tomorrow's toy



Then



Now

Electronic Toys

- Code complexity, memory arrays, analogue capabilities growing rapidly



- Processor word-width not primary cost driver
- RealView Development Suite 3.0: Thumb[®]-2 code density >3X better than 8051 on EEMBC
- RealView Microcontroller Development Kit supports MCUs out-of-the-box

- User interface, programmer's model, responsiveness crucial
 - Virtual machines improve time-to-market, simplify user programmability
 - Jazelle[®] DBX accelerates Java without start-up time or code bloat penalties



Electronic Toys

- Gaming and content are everywhere

- ARM7™ family processors decode 90 minutes of video in 32MB at 16MHz
- Swerve ES operates with any hardware accelerator (OpenGL ES 1.x driver) to deliver PS-1 quality
- Metro libraries and memories enable low-power and device portability



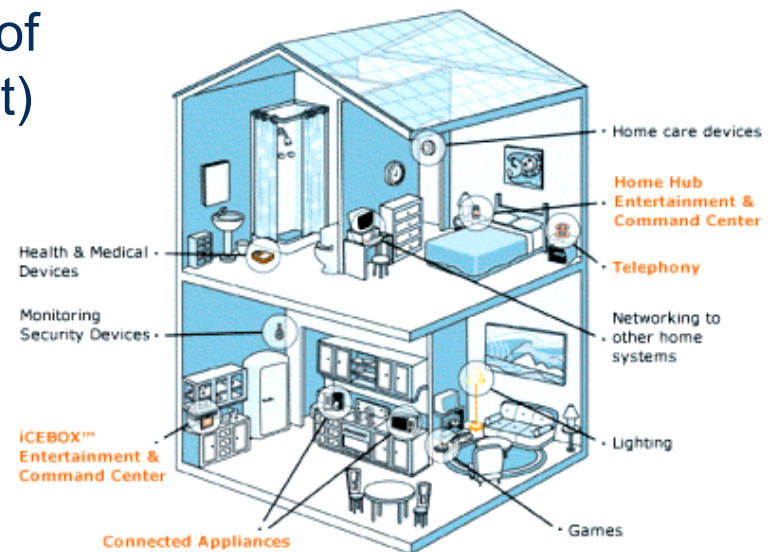
- “Education and Learning” or “Gaming and Content”?

- Nearly identical business models...
- Increased interactivity, behavioural aspects



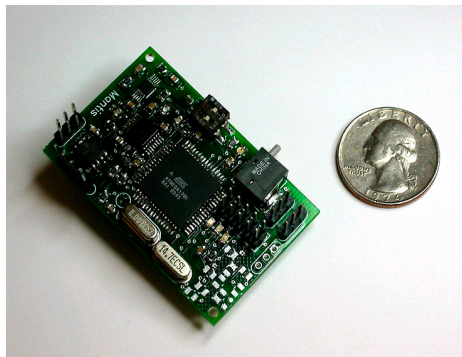
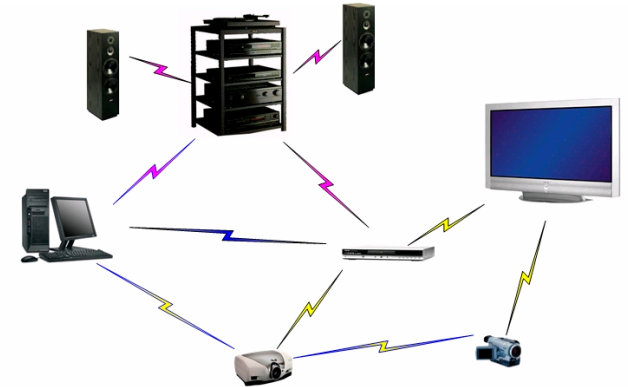
MCU-Class Networking

- Everyday devices getting smarter, more connected
 - Ambient intelligence
 - Pervasive computing
 - MCU utilisation of distributed information a “killer application”...but when?
 - Primary obstacles: complexity, ease of development, wired connectivity (cost)



MCU-Class Networking

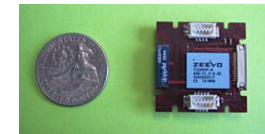
- Wireless standards emerging
 - IEEE 802.15.4 (RF PAN) in 2003 – Low power and low data rates
 - ZigBee – Remote monitoring and control
 - Z-Wave – Wireless home control
 - 6lowpan – IPv6 over low-power PAN
 - TinyOS – Event-based OS for wireless sensor networks
- Integration necessary to hit volume price points



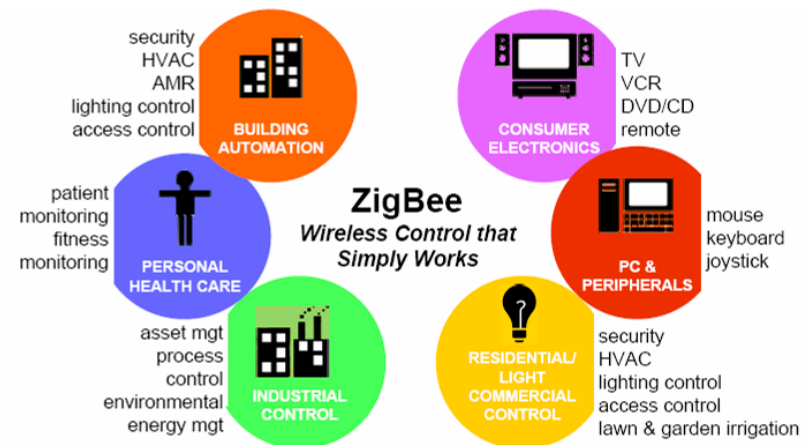
- Cortex-M3 processor capable of running complete software stacks
- Thumb-2 code density reduces memory array area costs
- RealView Microcontroller Development Kit provides consistency to maximise productivity

MCU-Class Networking

- Commercial applications leading the way
 - Machinery vibration monitoring:
Proactive maintenance
 - Commercial building automation:
Safety and efficiency
 - Homeland security:
Infrastructure and pipeline monitoring
 - Local data collection, processing saves RF power



- Ecosystem support imperative
 - Development cost roadblocks on the road to initial volume adoption
 - Challenges of headless debugging
 - Value realised in utilisation of data, not simply collection
 - Security, encryption challenges ever present

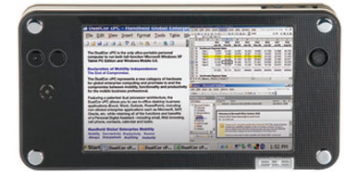


Emerging Category - Ultra Mobiles

- Three approaches
 - ARM processor-based Internet tablet
 - Nokia 770 Linux Tablet
 - Dual Processors
 - “VIA licenses ARM9™ family processors aimed for integration into next-generation comms and multimedia solutions”
 - Single or multiple ARM system with emulation



Nokia



DualCor Technologies
www.dualcor.com

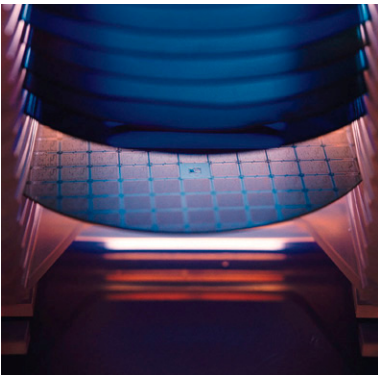


Flipstart

Access Library Program

- Selected ARM Physical IP available to design teams at no cost via web channel
 - Standard cells
 - Memory generators
 - I/O libraries
- Extensive foundry coverage: 1st Silicon, Chartered, DongbuAnam, Grace, HHNEC, HJTC, IBM, MagnaChip, Samsung, Silterra, SMIC, Tower, TSMC, UMC, Vanguard

World's Leading
IC Manufacturers



Physical IP



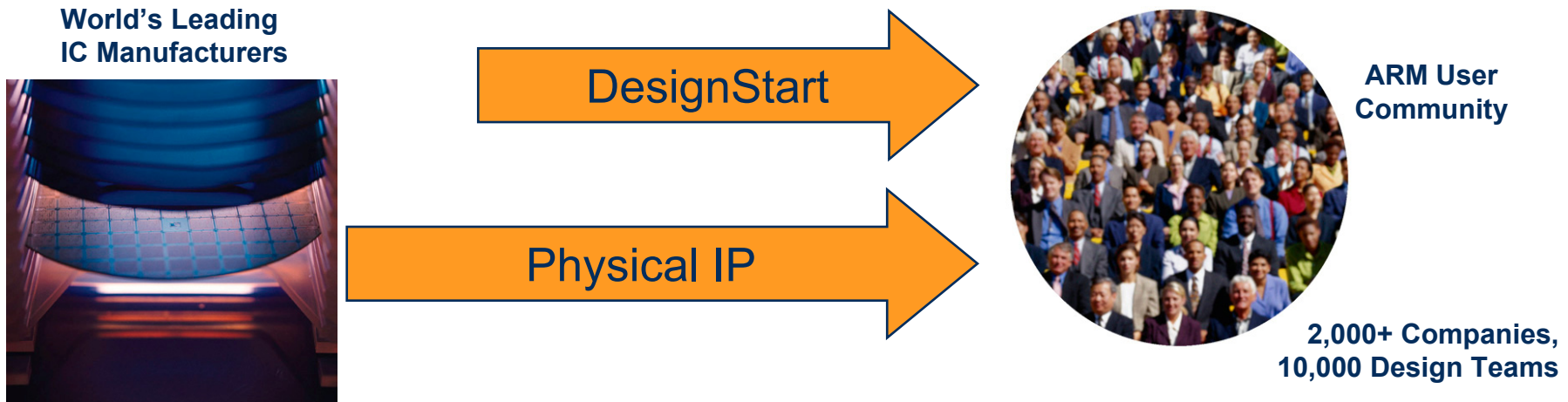
ARM User
Community



2,000+ Companies,
10,000 Design Teams

Leveraging DesignStart™

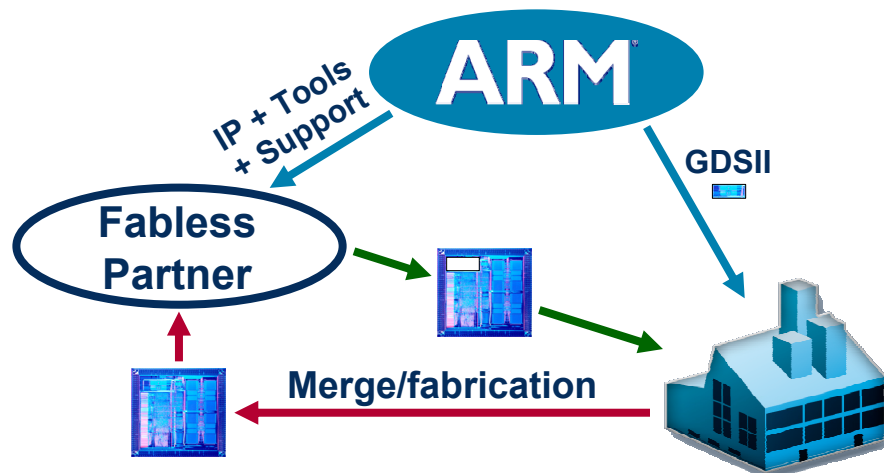
- ARM7TDMI® processor available directly to end users via web channel for easy evaluation through DesignStart program
- Leverages established Foundry Program, Artisan channel
- Greater IP evaluation choice at point of download
- Extends the ARM Partnership to a wider design community
- Flexible business models to help proliferate the ARM architecture and grow the ARM community



ARM Foundry Program

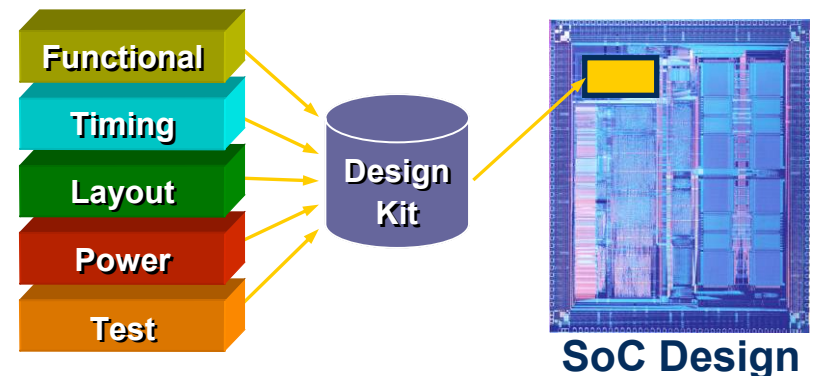
■ Foundry Program Model

- Simplified access to ARM technology for fabless semis
- Licensing flexibility to enable lower entry points
- Manufacturing choice with leading Foundries
- Design support network
- Secure transaction model

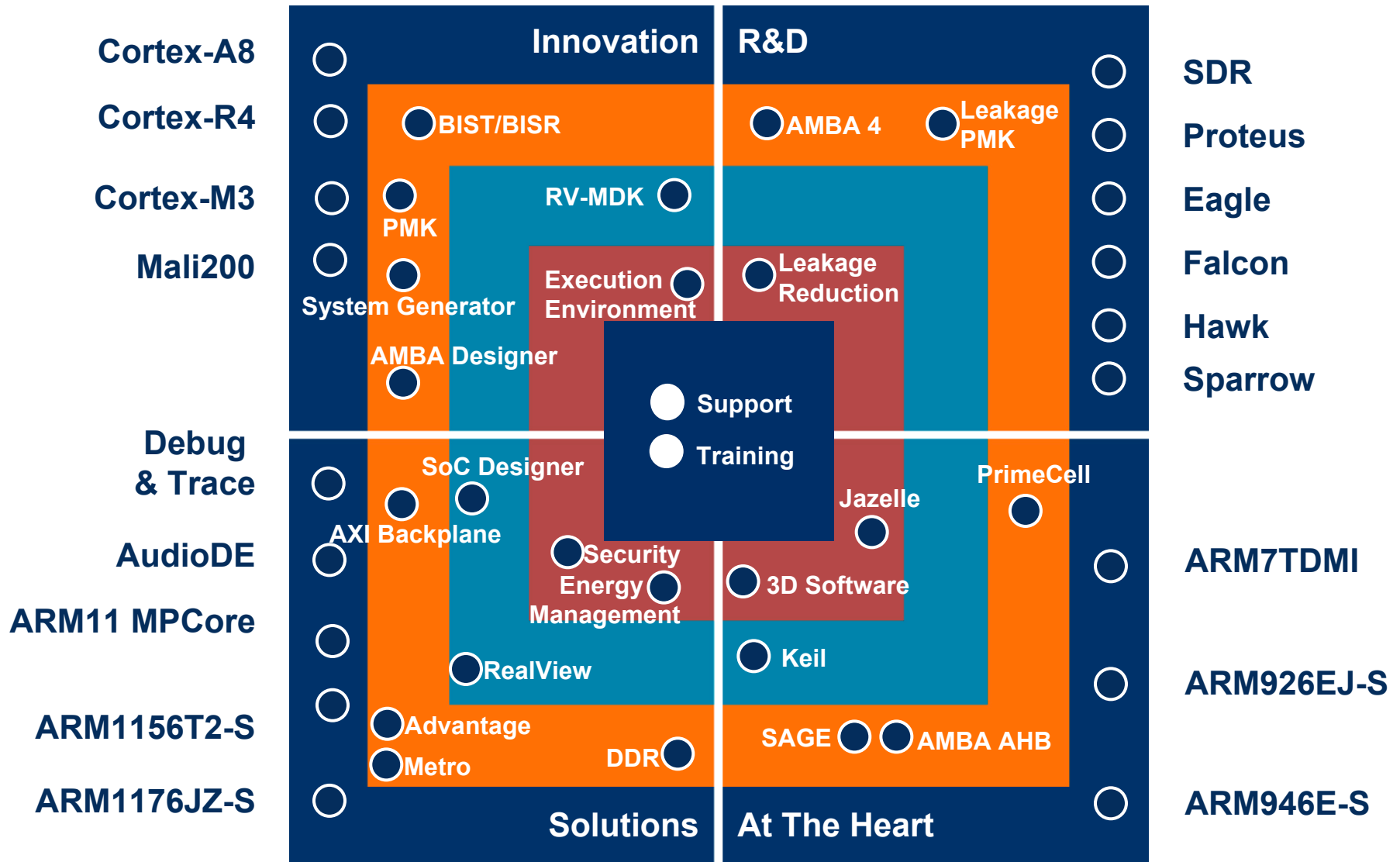


■ IP solutions

- Silicon-proven hard macros reduce time-to-market
- Complete “Black box” views for ease of integration
- AMBA Development Kit for fast-track design
- RealView Development Suite models for software simulation
- Roadmap to ARM11 family, 90nm

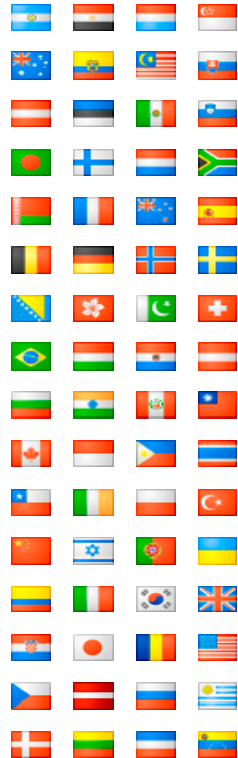


Emerging



Global Presence

- Different markets have different needs, preferences, channels, dynamics
 - GSMA Emerging Market Handset initiative
 - Brazilian National Microelectronics Plan, Dubai Silicon Oasis, Silicon Taiga
- Flexibility to address lower volume opportunities
 - ARM7 family processor (Actel) → EDN Hot 100 and Electronic Products 2005 Product of the Year
 - Proteus EAC in 2007
- Pricing without sacrifice
 - \$1 ARM MCU supported by RealView Microcontroller Dev Kit
 - Access Library program delivers memory generators, standard cell libraries, and I/Os for 15 foundries at no cost to design team
 - DesignStart program increases access to ARM7 family in 4 foundries



Conclusion

- Change is not a constant, it is accelerating
 - Silicon, software and sensor technology will revolutionise healthcare
 - Power now a primary constraint from implants to supercomputers
 - Toys, gaming, content becoming a personal experience
 - Utilisation of distributed information a killer application
 - Globalisation forcing new challenges and opportunities; creating new customers
- ARM is the ideal partner for your next application
 - ARM technology is relevant “outside the box” today, tomorrow
 - Leverage all cumulative investments to date
 - ARM is making ongoing, significant investments for the future

